Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let R be a relation on N defined by $x + 2y = 8$. The domain of R is

(A) $\{2, 4, 8\}$

(B) $\{2, 4, 6, 8\}$

(C) $\{2, 4, 6\}$

(D) $\{1, 2, 3, 4\}$

Q.2

Consider the following statements.
I. $\mathrm{If}(a, 1), (b, 2)$ and $(c, 1)$ are in $A \times B$ and $n(A) = 3, \mathrm{n(B)} = 2$, then $\mathrm{A} = \{\mathrm{a}, \mathrm{b}, \mathrm{c}\}$ and $\mathrm{B} = \{1, 2\}$
II. If $\mathrm{A} = \{1, 2\}$ and $\mathrm{B} = \{3, 4\}$, then $\mathrm{A} \times (\mathrm{B} \cap \phi)$ is equal to $\mathrm{A} \times \mathrm{B}$. Choose the correct option.

(A) Only I is true

(B) Only II is true

(C) Both are true

(D) Neither I nor II is true

Q.3

Let $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 3, 5, 7, 9\}$, which of the following is not relation from X to Y ?

(A) $R_1 = \{(x, y) : y = x + 2, x \in X, y \in Y\}$

(B) $R_2 = \{(1, 1), (2, 1), (3, 3), (4, 3), (5, 5)\}$

(C) $R_3 = \{(1, 1), (1, 3), (3, 5), (3, 7), (5, 7)\}$

(D) $R_4 = \{(1, 3), (2, 5), (2, 4), (7, 9)\}$

Q.4

Let $\mathrm{R} = \{(1, 3), (2, 2), (3, 2)\}$ and $\mathrm{S} = \{(2, 1), (3, 2), (2, 3)\}$ be two relations on set $\mathrm{A} = \{1, 2, 3\}$. Then $\mathrm{RoS} =$

(A) $\{(1, 3), (2, 2), (3, 2), (2, 1), (2, 3)\}$

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(B) $\{(3,2),(1,3)\}$

(C) $\{(2,3),(3,2),(2,2)\}$

(D) $\{(2,3),(3,2)\}$

Q.5

If R be a relation $<$ from $\mathrm{A}=\{1,2,3,4\}$ to $\mathrm{B}=\{1,3,5\}$ i.e., $(\mathrm{a}, \mathrm{b}) \in R \Leftrightarrow \mathrm{a}<\mathrm{b}$, then number of elements in RoR^{-1} is

(A) {(1,3),(2,5),(2,1),(4,5)}

(B) {(2,3),(3,1),(3,3),(5,4)}

(C) {(1,4),(2,3),(3,4),(5,5)}

(D) {(3, 3),(3, 5),(5, 3),(5, 5)}

Q.6

Let $\mathrm{E}=\{1,2,3,4\}$ and $\mathrm{F}=\{1,2\}$. Then the number of onto functions from E to Fis

(A) 12

(B) 11

(C) 13

(D) 14

Q.7

Let $\mathrm{A}=\{(\mathrm{n}, 2\mathrm{n}): \mathrm{n} \in \mathrm{N}\}$ and $\mathrm{B}=\{(2\mathrm{n}, 3\mathrm{n}): \mathrm{n} \in \mathrm{N}\}$. What is $A \cap B$ equal to ?

(A) $\{(\mathrm{n}, 6\mathrm{n}): \mathrm{n} \in \mathrm{N}\}$

(B) $\{(2n, 6n): n \in N\}$

(C) $\{(n, 3n): n \in N\}$

(D) ϕ

Q.8

If $aN=\{ax: x \in \mathbf{N}\}$ and $bN \cap cN=dN$, where $b, c \in N$ are relatively prime, then

(A) $\mathrm{d}=\mathrm{bc}$

(B) $\mathrm{c}=\mathrm{bd}$

(C) $\mathrm{b}=\mathrm{cd}$

(D) None of these

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Q.9

In a class of 55 students, the number of students studying different subjects are 23 in Mathematics, 24 in Physics, 19 in Chemistry, 12 in Mathematics and Physics, 9 in Mathematics and Chemistry, 7 in Physics and Chemistry and 4 in all the
three subjects. The number of students who have taken exactly one subject is

(A) 6

(B) 9

(C) 7

(D) 22

Q.10

A set A has 3 elements and another set B has 6 elements.
Then

(A) $3 \leq \mathrm{n}(\mathrm{A} \cup \mathrm{B}) \leq 6$

(B) $3 \leq \mathrm{n}(\mathrm{A} \cup \mathrm{B}) \leq 9$

(C) $6 \leq \mathrm{n}(\mathrm{A} \cup \mathrm{B}) \leq 9$

(D) $0 \leq \mathrm{n}(\mathrm{A} \cup \mathrm{B}) \leq 9$

Q.11

Let X and Y be two non-empty sets such that $X \cap A = Y \cap A = \phi$ and $X \cup A = Y \cup A$ for some non-empty set A. Then

(A) X is a proper subset of Y

(B) Y is a proper subset of X

(C) $X = Y$

(D) X and Y are disjoint sets

Q.12

In a town of 10,000 families, it was found that 40% families buy newspaper A, 20% families buy newspaper B and 10% families buy newspaper C. 5% families buy A and B, 3% buy B and C and 4% buy A and C. If 2% families buy all the
newspapers, then

(A) 3,300 families buy A only

(B) 1,400 families buy B only.

(C) 4000 families buy none of A, B and C

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(D) All are correct

Q.13

Let $n(\mathrm{U})=700, n(\mathrm{A})=200, n(\mathrm{B})=300, n(\mathrm{A} \cap \mathrm{B})=100$, then $n\left(\mathrm{A}' \cap \mathrm{B}'\right)$ is equal to

(A) 400

(B) 600

(C) 300

(D) None of these

Q.14

In a class of 80 students numbered from 1 to 80, all odd numbered students opt of Cricket, students whose numbers are divisible by 5 opt for Football and those whose numbers are divisible by 7 opt for Hockey. The number of students who do not opt any of the three games, is

(A) 13

(B) 24

(C) 28

(D) 52

Q.15

The set $(A \backslash B) \cup (B \backslash A)$ is equal to

(A) $[A \backslash (A \cap B)] \cap [B \backslash (A \cap B)]$

(B) $(A \cup B) \backslash (A \cap B)$

(C) $A \backslash (A \cap B)$

(D) $\overline{A \cap B \backslash A \cup B}$

Q.16

If A is the set of the divisors of the number 15, B is the set of prime numbers smaller than 10 and C is the set of even numbers smaller than 9, then $(A \cup C) \cap B$ is the set

(A) $\{1,3,5\}$

(B) $\{1,2,3\}$

(C) $\{2,3,5\}$

(D) $\{2,5\}$

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Q.17

Two finite sets have m and n elements. The number of subsets of the first set is 112 more than that of the second set. The values of m and n are, respectively,

(A) 4,7

(B) 7,4

(C) 4,4

(D) 7,7

Q.18

If $n(\mathrm{A})=1000, n(\mathrm{B})=500$ and if $n(\mathrm{A} \cap \mathrm{B}) \geq 1$ and $n(\mathrm{A} \cup \mathrm{B})=p$, then

(A) $500 \leq p \leq 1000$

(B) $1001 \leq p \leq 1498$

(C) $1000 \leq p \leq 1498$

(D) $1000 \leq p \leq 1499$

Q.19

Let A, B, C be finite sets. Suppose that $n(A)=10, n(B)=15, n(\mathrm{C})=20, \mathrm{n}(\mathrm{A} \cap \mathrm{B})=8$ and $\mathrm{n}(\mathrm{B} \cap \mathrm{C})=9$. Then the possible value of $n(A \cup B \cup C)$ is

(A) 26

(B) 27

(C) 28

(D) Any of the three values $26, 27, 28$ is possible

Q.20

Let $\mathrm{A}=\{\mathrm{x} \in \mathrm{W}$, the set of whole numbers and $x<3\}, B=\{x \in N$, the set of natural numbers and $2 \leq x<4\}$ and $C=\{3,4\}$, then how many elements will $(\mathrm{A} \cup \mathrm{B}) \times \mathrm{C}$ contain?

(A) 6

(B) 8

(C) 10

(D) 12

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Q.1 Answer:

$\{2, 4, 6\}$

Solution:

R be a relation on N defined by $x + 2y = 8$

$\therefore \quad \mathrm{R} = \{(2, 3); (4, 2); (6, 1)\}$

Hence, Domain of $R = \{2, 4, 6\}$

Q.2 Answer:

Only I is true

Solution:

I. $A =$ Set of first elements $= \{a, b, c\}$ $\mathrm{B} =$ Set of second elements $= \{1, 2\}$

II. $\mathrm{B} \cap \phi = \phi \quad \therefore \mathrm{A} \times \phi = \phi$

Q.3 Answer:

$R_4 = \{(1, 3), (2, 5), (2, 4), (7, 9)\}$

Solution:

R_1 is a relation from X to Y because $R_1 \subseteq X \times Y$. R_2 is a relation from X to Y because $R_2 \subseteq X \times Y$. R_3 is a relation from X to Y because $R_3 \subseteq X \times Y$. R_4 is not a relation from X to Y because $(2, 4), (7, 9) \notin X \times Y$

Q.4 Answer:

$\{(2, 3), (3, 2), (2, 2)\}$

Solution:

Here $\mathrm{R} = \{(1, 3), (2, 2), (3, 2)\}, \mathrm{S} = \{(2, 1), (3, 2), (2, 3)\}$

Then $\mathrm{RoS} = \{(2, 3), (3, 2), (2, 2)\}$

Q.5 Answer:

{(3, 3),(3, 5),(5, 3),(5, 5)}

Solution:

We have, $\mathrm{R} = \{(1, 3), (1, 5), (2, 3), (2, 5), (3, 5), (4, 5)\}$

$\mathrm{R}^{-1} = \{(3, 1), (5, 1), (3, 2), (5, 2), (5, 3), (5, 4)\}$

Hence $\mathrm{RoR}^{-1} = \{(3, 3), (3, 5), (5, 3), (5, 5)\}$

Q.6 Answer:

14

Solution:

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If set A has m elements and set B has n elements then number of onto functions from A to B is $\sum_{r=1}^{n}(-1)^{n-r}$ ${}^{n}C_{r}r^{m}$ where $1 \leq n \leq m$

Here $E=\{1,2,3,4\}, \quad F=\{1,2\}; m=4, n=2$

$\therefore$ no. of onto functions from E to F

$=\sum_{r=1}^{2}(-1)^{2-r}$

${}^{2}C_{r}(r)^{4}=(-1)^{2}C_{1}+{}^{2}C_{2}(2)^{4}$

$=-2+16=14$

Q.7 Answer:

ϕ

Solution:

$A=\{(n, 2n): n \in N\}$ and $B=\{(2n, 3n)\}: n \in N$

Listing few members of each set $A=\{(1,2),(2,4),(3,6), \ldots .\}$

$B=\{(2,3),(4,6),(6,9) \ldots \ldots .\}$

There is no member common to both these sets, hence. $A \cap B=\phi$

Q.8 Answer:

$d=bc$

Solution:

$bN=\{bx: x \in N\}$

$cN=\{cx: x \in N\}$

$\therefore bN \cap cN=\{x: x$ is multiple of b and c both $\}$

$=\{x: x$ is multiple of l.c.m. of b and $c\}=\{x: x$ is multiple of $bc\}$

[given b and c are relatively prime $\therefore$ l.c.m. of b and $c=bc$]

$\therefore bN \cap cN=\{bcx: x \in N\}=dN$ (Given)

$\therefore d=bc$

Q.9 Answer:

22

Solution:

$n(M)=23, n(P)=24, n(C)=19$

$n(M \cap P)=12, n(M \cap C)=9, n(P \cap C)=7$

$n(M \cap P \cap C)=4$

We have to find $n\left(M \cap P' \cap C'\right), n\left(P \cap M' \cap C'\right)$,

$n\left(C \cap M' \cap P'\right)$

Now $n\left(M \cap P' \cap C'\right)=n\left[M \cap (P \cup C)'\right]$

$=n(M)-n[(M \cap (P \cup C)]$

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$= n(M) - n[(M \cap P) \cup (M \cap C)]$
$= n(M) - n(M \cap P) - n(M \cap C) + n(M \cap P \cap C)$
$= 23 - 12 - 9 + 4 = 27 - 21 = 6$
$n\left(P \cap M \cap C'\right) = n\left[P \cap (M \cup C)'\right]$
$= n(P) - n[P \cap (M \cup C)]$
$= n(P) - n[(P \cap M) \cup (P \cap C)]$
$= n(P) - n(P \cap M) - n(P \cap C) + n(P \cap M \cap C)$
$= 24 - 12 - 7 + 4 = 9$
$n\left(C \cap M' \cap P'\right)$
$= n(C) - n(C \cap P) - n(C \cap M) + n(C \cap P \cap M)$
$= 19 - 7 - 9 + 4 = 23 - 16 = 7$
$\therefore$ **Total** $= 6 + 9 + 7 = 22$

Q.10 Answer:

$6 \leq \mathrm{n}(\mathrm{A} \cup \mathrm{B}) \leq 9$

Solution:

We have $\min n(A \cup B) = \max\{n(A), n(B)\} = \max\{3, 6\} = 6$
$\max n(A \cup B) = n(A) + n(B) = 9$
$\therefore 6 \leq \mathrm{n}(\mathrm{A} \cup \mathrm{B}) \leq 9$

Q.11 Answer:

$X = Y$

Solution:

Suppose a $\in X$ **and** $a \in A \Rightarrow a \in X \cup A \Rightarrow a \in Y \cup A$
$\Rightarrow a \in Y$ **and** $a \in A (\because X \cup A = Y \cup A)$
$\Rightarrow a \in Y \cap A \Rightarrow Y \cap A$ **is non-empty**
This contradicts that $Y \cap A = \phi$ **So,** $X = Y$

Q.12 Answer:

All are correct

Solution:

Let P **= set of families buying** A, Q **= set of families buying** B **and** R **= set of families buying** C.
$\therefore \mathrm{n}(\mathrm{P}) = 40\%$ **of** $10,000 = 4,000$, **similarly**
$\mathrm{n}(\mathrm{Q}) = 2,000, \mathrm{n}(\mathrm{R}) = 1,000$
$\mathrm{n}(\mathrm{P} \cap \mathrm{Q}) = 500, \mathrm{n}(\mathrm{Q} \cap \mathrm{R}) = 300$
$\mathrm{n}(\mathrm{P} \cap \mathrm{R}) = 400$ **and** $\mathrm{n}(\mathrm{P} \cap \mathrm{Q} \cap \mathrm{R}) = 200$
(i) Number of families buying only $\mathrm{A} = \mathrm{n}\left(\mathrm{P} \cap \mathrm{Q}' \cap \mathrm{R}'\right)$
$= \mathrm{n}\left(\mathrm{P} \cap (\mathrm{Q} \cup \mathrm{R})'\right) = \mathrm{n}(\mathrm{P}) - \mathrm{n}(\mathrm{P} \cap (\mathrm{Q} \cup \mathrm{R}))$
$= \mathrm{n}(\mathrm{P}) - [\mathrm{n}(\mathrm{P} \cap \mathrm{Q}) + \mathrm{n}(\mathrm{P} \cap \mathrm{R}) - \mathrm{n}((\mathrm{P} \cap \mathrm{Q}) \cap (\mathrm{P} \cap \mathrm{R}))]$

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$= \mathrm{n(P)} - \mathrm{n(P \cap Q)} - \mathrm{n(P \cap R)} + \mathrm{n(P \cap Q \cap R)}.$
$= 4,000 - 500 - 400 + 200 = 3,300$

(ii) Number of families buying only B $= \mathrm{n(Q)} - \mathrm{n(P \cap Q)} - \mathrm{n(Q \cap R)} + \mathrm{n(P \cap Q \cap R)}$ $[\mathrm{see(i)}]$
$= 2,000 - 500 - 300 + 200 = 1,400.$

(iii) Number of families buying none of $\mathrm{A, B}$ **anc** $\mathrm{C} = \mathrm{n}\left(\mathrm{P}' \cap \mathrm{Q}' \cap \mathrm{R}'\right) = \mathrm{n}\left(\mathrm{P}' \cap (\mathrm{Q} \cup \mathrm{R})'\right)$
$= \mathrm{n}\left\{(\mathrm{P} \cup (\mathrm{Q} \cup \mathrm{R}))'\right\} = 10000 - \mathrm{n(P \cup Q \cup R)}$
$= 10,000 - [\mathrm{n(P)} + \mathrm{n(Q)} + \mathrm{n(R)} - \mathrm{n(P \cap Q)}$
$- \mathrm{n(Q \cap R)} - \mathrm{n(P \cap R)} + \mathrm{n(P \cap Q \cap R)}]$
$= 10,000 - [4,000 + 2,000 + 1,000 - 500 - 300 - 400 + 200]$
$= 10,000 - 6,000 = 4,000.$

Note : For sets A, B**, we have**
$(\mathrm{A \cap B}) \cup (\mathrm{A \cap B}') = \mathrm{A} \cap (\mathrm{B} \cup \mathrm{B}') = \mathrm{A} \cap \mathrm{U} = \mathrm{A}$
and $(A \cap B) \cap (A \cap B') = A \cap (B \cap B') = A \cap \phi = \phi$
$\therefore n(A) = n(A \cap B) + n\left(A \cap B'\right)$ **or** $n\left(A \cap B'\right)$
$= n(A) - n(A \cap B)$
Replacing A **by** P **and** B **by** $Q \cup R$**, we have** $\mathrm{n}\left(\mathrm{P} \cap (\mathrm{Q} \cup \mathrm{R})'\right) = \mathrm{n(P)} - \mathrm{n}(\mathrm{P} \cap (\mathrm{Q} \cup \mathrm{R}))$
etc.

Hence all options are correct.

Q.13 Answer:

300

Solution:

$n\left(\mathrm{A}' \cap \mathrm{B}'\right) = n(\mathrm{A} \cup \mathrm{B})' = n(\mathrm{U}) - n(\mathrm{A} \cup \mathrm{B})$
$= n(\mathrm{U}) - [n(\mathrm{A}) + n(\mathrm{B}) - n(\mathrm{A} \cap \mathrm{B})]$
$= 700 - [200 + 300 - 100] = 300$

Q.14 Answer:

28

Solution:

Numbers which are divisible by 5 are $5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80$ **they are 16 in numbers. Now, Numbers which are divisible by 7 are**
$7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77$ **they are 11 in numbers. Also, total odd numbers** $= 40$
Let C **represents the students who opt. for cricket,** F **for football and** H **for hockey.** $\therefore$ **we have**
$n(C) = 40, n(F) = 16, n(H) = 11$
Now, $C \cap F =$ **Odd numbers which are divisible by 5 .** $C \cap H =$ **Odd numbers which are divisible by** 7**.** $F \cap H =$ **Numbers which are divisible by both 5 and 7** $n(C \cap F), 8, n(C \cap H) = 6$
$n(F \cap H) = 2, \mathrm{n}(C \cap F \cap H) = 1$

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We Know $n(\mathrm{C} \cup F \cup H) = n(C) + n(F) + n(H)$
$-n(C \cap F) - n(C \cap H)$
$-n(F \cap H) + n(C \cap H \cap F)$
$n(\mathrm{C} \cup \mathrm{F} \cup \mathrm{H}) = 67 - 16 + 1 = 52$
$\therefore n\left(C' \cap F' \cap H'\right)$
$=$ **Total students** $-n(C \cup F \cup H)$
$n\left(C' \cap F' \cap H'\right) = 80 - 52 = 28$

Q.15 Answer:

$(A \cup B) \backslash (A \cap B)$

Solution:

Given set can be written as $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
(By definition of symmetric difference) Hence, $(\mathrm{A} \backslash \mathrm{B}) \cup (\mathrm{B} \backslash \mathrm{A}) = (\mathrm{A} \cup \mathrm{B}) \backslash (\mathrm{A} \cap \mathrm{B})$

Q.16 Answer:

$\{2, 3, 5\}$

Solution:

$\mathrm{A} = \{1, 3, 5, 15\}, \mathrm{B} = \{2, 3, 5, 7\} \mathrm{C} = \{2, 4, 6, 8\}$
$\therefore \mathrm{A} \cup \mathrm{C} = \{1, 2, 3, 4, 5, 6, 7, 8, 15\}$
$(\mathrm{A} \cup \mathrm{C}) \cap \mathrm{B} = \{2, 3, 5\}$

Q.17 Answer:

7,4

Solution:

$2^m - 2^n = 112 \Rightarrow 2^n \left(2^{m-n} - 1\right) = 16.7$
$\therefore \quad 2^n \left(2^{m-n} - 1\right) = 2^4 \left(2^3 - 1\right)$
Comparing we get $n = 4$ **and** $m - n = 3 \Rightarrow n = 4$ **and** $m = 7$

Q.18 Answer:

$1000 \leq p \leq 1499$

Solution:

$\mathrm{n}(\mathrm{A}) = 1000, \mathrm{n}(\mathrm{B}) = 500, \mathrm{n}(\mathrm{A} \cap \mathrm{B}) \geq 1,$
$\mathrm{n}(\mathrm{A} \cup \mathrm{B}) = \mathrm{p}$
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$p = 1000 + 500 - n(A \cap B)$
$1 \leq n(A \cap B) \leq 500$
Hence $\mathrm{p} \leq 1499$ **and** $\mathrm{p} \geq 1000$
$1000 \leq \mathrm{p} \leq 1499$

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Q.19 Answer:

Any of the three values $26, 27, 28$ is possible

Solution:

We have $n(A \cup B \cup C) = n(A) + n(B) + n(C) -$
$n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$
$= 10 + 15 + 20 - 8 - 9 - \mathrm{n(C \cap A)} + \mathrm{n(A \cap B \cap C)}$
$= 28 - \{\mathrm{n(C \cap A)} - \mathrm{n(A \cap B \cap C)}\}$
...(i) Since $n(C \cap A) \geq n(A \cap B \cap C)$
We have $n(C \cap A) - n(A \cap B \cap C) \geq 0$
(ii) From (i) and (ii)
$n(A \cup B \cup C) \leq 28$
(iii) Now, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 10 + 15 - 8 = 17$
and $\mathrm{n(B \cup C)} = \mathrm{n(B)} + \mathrm{n(C)} - \mathrm{n(B \cap C)}$
$= 15 + 20 - 9 = 26$
Since, $n(A \cup B \cup C) \geq n(A \cup C)$ **and**
$n(A \cup B \cup C) \geq n(B \cup C)$**, we have**
$n(A \cup B \cup C) \geq 17$ **and** $n(A \cup B \cup C) \geq 26$
Hence $n(A \cup B \cup C) \geq 26$
From (iii) and (iv) we obtain $26 \leq n(A \cup B \cup C) \leq 28$
Also $\mathrm{n(A \cup B \cup C)}$ **is a positive integer** $\therefore \mathrm{n(A \cup B \cup C)} = 26$ **or 27 or 28**

Q.20 Answer:

8

Solution:

We have $\mathrm{A} = \{0, 1, 2\}$
$\mathrm{B} = \{2, 3\}$
$\mathrm{C} = \{3, 4\}$
$\mathrm{A \cup B} = \{0, 1, 2, 3\}$
$(\mathrm{A \cup B}) \times \mathrm{C} = \{(0, 3), (0, 4), (1, 3); (1, 4); (2, 3)$
$(2, 4), (3, 3); (3, 4)\}$
$\therefore \mathrm{n}[(\mathrm{A \cup B}) \times \mathrm{C}] = 8$

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